Strict Splitting Counts in Reconstructed Elliptic-Curve Candidate Streams: An Exact Reproduction and Its Limits

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Abstract

We reproduce a previously reported arithmetic separation between published elliptic curves and a particular set of unpublished candidates reconstructed from historical seed streams. For each curve over a prime field, we count odd primes at most 691 for which the Frobenius discriminant is a nonzero square. All eleven unpublished candidates have larger counts than every publication in their respective field, across four fields containing eight publications. Two further publications, over the 384- and 521-bit fields, are retained without unpublished same-field controls. Separate GMP and PARI implementations agree on 6,342 odd-prime characters, 303 cutoff positions and all 4,480 label assignments preserving publication counts within streams. Exactly 23 assignments separate at some cutoff in the declared family, reproducing the prior report. Including zero characters destroys the separation at 691 in the 192-bit field. The cutoff, labels and reported outcome were known before this replay: its contribution is computational reproduction, explicit convention and cohort analysis, and an auditable evidence record. The result neither authenticates a historical selection rule nor establishes a discrete-logarithm advantage.

Keywords: elliptic curves; Frobenius discriminant; Elkies primes; parameter generation; reproducibility; selection bias.

1 Introduction

The reconstruction of timestamp-initialized seed streams for historically published NIST and ANSI elliptic curves leaves a separate question: why were some other reconstructed candidates not published? Public accounts identify an additional unexplained selection step after their stated arithmetic screens [2, 9]. Recovering a seed stream and reconstructing an eligible candidate do not, by themselves, show that the original program visited that candidate or rejected it for a particular reason.

This paper examines an observation reported earlier by the account asanso [3]: at a small-prime cutoff of 691, published curves have fewer strictly split primes than unpublished candidates in the same field. We reproduce that observation on its fixed 21-choice cohort. We do not claim discovery of the observation. The reported cutoff, outcomes and labels were visible before the computational protocol was frozen.

Our contributions are an exact replay with a separate implementation, a complete row-level account of the compared choices, reproduction of the specified finite label calibration, and a sensitivity analysis separating nonzero splitting from zero characters. We explain how this property relates to generator/coefficient comparisons and identify the historical and security conclusions that

remain unsupported. No new point counting, discrete-logarithm attack, runtime comparison or isogeny construction is performed for this manuscript.

The paper is a replication note prepared from committed evidence [7]. All displayed counts, indices, cutoffs and thresholds are decimal. Parameter files retain their original explicit integer encodings.

2 Arithmetic setting and precise statistic

2.1 Frobenius discriminant

Let $E/\mathbb{F}_p$ be one of the nonsingular prime-field curves in the cohort. The recorded full group order is $n = \#E(\mathbb{F}_p)$, including the identity; the supplied cofactors are one. Set

$$t = p + 1 - n, \qquad \Delta = t^2 - 4p. \tag{1}$$

For an odd prime $\ell \neq p$, let $\chi_\ell(E) = (\Delta/\ell) \in \{-1, 0, +1\}$ be the Legendre symbol. The Frobenius characteristic polynomial modulo ℓ is $X^2 - tX + p$. A positive discriminant character gives two distinct roots and two Frobenius-stable cyclic subgroup directions, hence two corresponding degree-ℓ isogeny directions defined over $\mathbb{F}_p$. This is standard point-counting structure [8]. We compute the characters, not the isogeny maps.

Define the *strict splitting count*

$$S_L(E) = \#\{\ell : 3 \leq \ell \leq L,\ \ell \text{ prime},\ \chi_\ell(E) = +1\}. \tag{2}$$

Separately record the zero-character and inclusive counts

$$R_L(E) = \#\{\ell : 3 \leq \ell \leq L,\ \ell \text{ prime},\ \chi_\ell(E) = 0\}, \qquad I_L(E) = S_L(E) + R_L(E). \tag{3}$$

The term *Elkies prime* alone does not specify which count is meant: definitions using a square discriminant can include zero. We use Equation (2) for the primary result and keep the inclusive statistic separate. A zero character is not evidence that rational isogenies are absent.

The prime 2 is excluded from these definitions. In the recorded cutoff enumeration it is a zero-count boundary, with its character explicitly unavailable under the odd-prime convention. Every cohort characteristic is larger than the maximum cutoff 2,000, so no contributing prime equals p.

2.2 Within-field separation

Let $\mathcal{P}_f$ be the publications and $\mathcal{U}_f$ the compared unpublished choices in field f. The fixed lower-count direction separates at cutoff L precisely when

$$\max_{E \in \mathcal{P}_f} S_L(E) < \min_{E \in \mathcal{U}_f} S_L(E) \quad \text{for every field containing both labels.} \tag{4}$$

Thus each paired field has an upper threshold accepting every publication and excluding every compared unpublished candidate. Ties prevent separation. Thresholds follow the observed labeled envelopes; they are not authenticated historical constants. Fields containing only publications remain in the census but do not validate discrimination.

3 Cohort, identities and historical scope

The cohort contains ten publications and eleven unpublished reconstructed candidates across six streams. The 192- and 239-bit streams each have three publications; the 224- and 256-bit streams each have one publication with unpublished controls. The 384- and 521-bit streams each supply one publication without such controls. The 239-bit entries are ANSI examples; we do not use P-239 as a NIST curve name.

A row key such as `p192-139` gives the field size and stream position. It is a local record key, not a new standard name or an isomorphism claim. The byte-preserved `data/inputs.tsv` fixes every row by stream, index, exact p, b, n and seed, with $a = -3 \bmod p$. Historical publication names are in Appendix A. The original four supplied models have frozen canonical ICV1 records in the earlier assessment identified by our data manifest. No new active catalogue alias is introduced.

Ascertainment of the eleven. These eleven are a retrospective comparison set, not a proved complete set of historical rejections. An earlier audit of the November 1997 X9.62 draft found 69 seed positions with certified full-prime-order branches meeting its listed numerical conditions: ten published and 59 unpublished [1, 5]. The eleven form a subset under additional root and order-width conventions. That broader result is inherited evidence, not recomputed here; its full-prime-order census is also only a lower bound on the draft's near-prime acceptance space.

The present separator therefore does not classify all 59 unpublished positions, or every curve the original software might have examined. No such extension was tested here. Historical use requires a justified link between the actual implementation and this restricted cohort, or independent predictions beyond it. We use *unpublished* to describe a recorded publication outcome, not observed visitation or rejection.

Conventions and missing data. The certified orders are inherited, not proved anew. The two P-224 coefficient representatives examined earlier preserve the same order and hence this statistic. All eleven historical unpublished generators remain unavailable. Assigned comparison points do not substitute for historical observations.

4 Reproduction method

4.1 Two implementations

The native producer uses GMP integers, trial division for the small primes and enumerated quadratic-residue tables. The separate verifier uses PARI prime enumeration and Kronecker symbols. The producer enumerates bitmasks; the verifier uses recursive label combinations and direct within-field pair comparisons. Both reconstruct all six output files, including the summary. Underlying GMP integer primitives may be shared; independence here concerns the implementations, not independent integer libraries.

There are 302 odd primes through 2,000. Across 21 choices the programs verify 6,342 odd-prime characters. They retain 303 prime cutoff positions, including the exceptional boundary at 2. That boundary is not silently converted to an ordinary character observation. Unavailable generators and incomplete historical records are not numeric labels.

4.2 Finite label calibration

The enumeration preserves publication counts in each original stream:

$$\binom{5}{3}\binom{4}{1}\binom{8}{3}\binom{2}{1}\binom{1}{1}\binom{1}{1} = 4,480. \tag{5}$$

Every assignment receives the same 303 cutoffs and fixed lower-count direction from Equation (4). Field thresholds follow that assignment's publication envelope, as for the observed labels. Singleton labels remain fixed. Strict and inclusive counts form separate search families.

This is finite combinatorial calibration. A historical randomization-test interpretation would also require exchangeability assumptions that are not established. The enumeration accounts for the declared cutoff search, not every trait, direction, coefficient convention or multivariate rule explored elsewhere.

4.3 Execution and refusal controls

The protocol was frozen before the replay at source `077c41e7d414cf72faf20bf9b474b7c35dae7bd8`, with digest

`eeb7482bcda69477a360a7c28749b7f737d377dbe3298bec94346e4efbcdbbb7`.

This ordering protects the replay record; it does not make the externally chosen hypothesis prospective.

Six phases were planned, attempted and completed in one actual attempt. The positive control, producer and independent verifier each exited zero. Deliberately corrupted character, calibration and summary outputs were each refused with the required native exit one, identifying the intended mismatching file. These are correct refusals, not successful verification of corrupted data. Canonical export admission with file verification reports zero findings.

Address space was limited uniformly to 1536 MiB. Watchdogs were 30 seconds for the positive and each refusal control, and 120 seconds for production and verification. These are resource limits, not measured scientific runtimes. The uploaded checkpoint gate passed before execution. There was no retry or restore of the actual scientific attempt. An earlier planning invocation failed before any control ran because an external timing utility was absent; the retained second invocation used shell timing. The local exact-analysis fallback and cloud-tool limitations are documented in the original record.

5 Observed separation

Table 1 reports exact values, not sample means with sampling intervals. The upper envelopes are 60, 50, 57 and 57 for the four paired fields. Every compared unpublished choice is above its field's envelope. Figure 1 displays all 21 rows.

A global threshold cannot fit these labels: published 192-bit count 60 exceeds unpublished 224-bit count 53. The 256-bit pair differs by one count, an exact distinction rather than a demonstrated large security effect. Across the complete cutoff family, the labels separate at exactly 691, 701, 709 and 769. These correlated cumulative counts are not independent historical confirmations.

Field	Published S_{691}	Unpublished S_{691}	Upper bound
192-bit	57, 58, 60	61, 62	60
224-bit	50	53, 56, 59	50
239-bit ANSI	42, 57, 57	61, 63, 66, 67, 71	57
256-bit	57	58	57
384-bit	54	no control	not tested
521-bit	52	no control	not tested

Table 1: Strict counts at decimal cutoff 691 and observed within-field publication envelopes. Singleton publications do not validate a rule. Source: native `data/at691.tsv`.

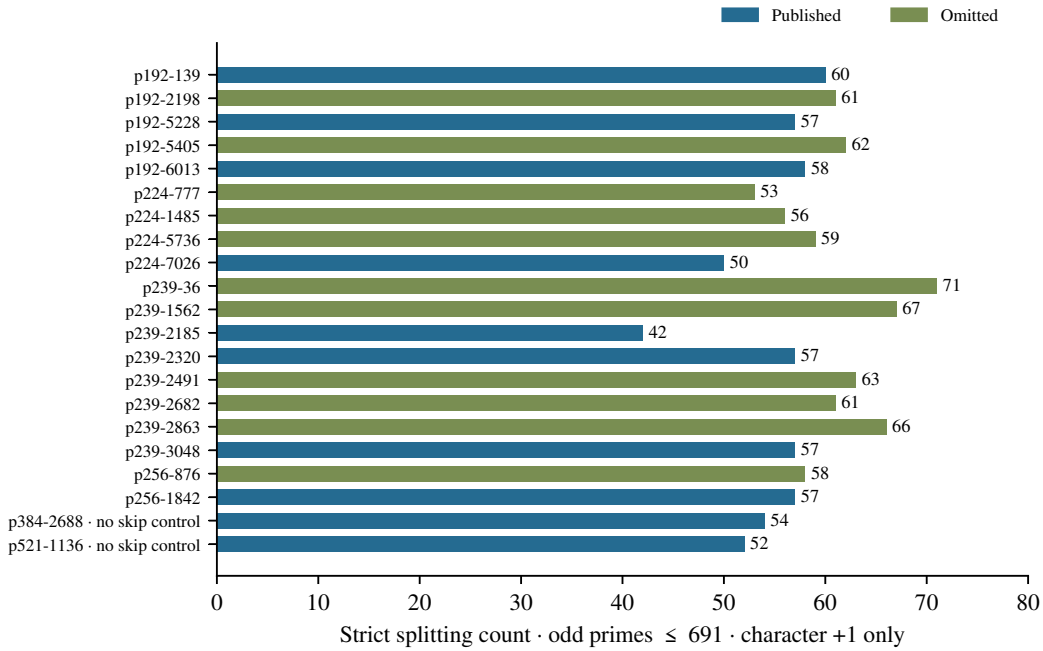


Figure 1: Every original choice, including singleton publications. This vector rendering reuses the frozen recorded figure. Values are exact S_{691} counts, not attack costs or security scores.

Index	Outcome	S_{691}	R_{691}	I_{691}
139	NIST P-192	60	3	63
2198	skipped	61	0	61
5228	published	57	1	58
5405	skipped	62	0	62
6013	published	58	1	59

Table 2: The complete 192-bit subcohort. Zero characters change NIST P-192 from 60 to 63 under inclusive counting.

5.1 P-192 and zero characters

NIST P-192 is the publication at index 139. Its strict count is 60, whereas unpublished indices 2198 and 5405 have counts 61 and 62. The other publications have counts 57 and 58. P-192 has three zero-character primes, making its inclusive count 63. The unpublished counts remain 61 and 62. An inclusive upper envelope accepting every publication therefore also accepts those candidates: the cutoff-691 separation fails. Zero characters cannot be interpreted as absent isogenies or removed without specifying the convention.

6 Calibration and inference

Exactly 23 of the 4,480 permitted assignments have a separating cutoff in the strict family, reproducing the prior report [3, 7]. At fixed cutoff 691, one assignment separates. The smaller fixed-cutoff fraction is not a new confirmatory probability: 691 was selected retrospectively. Freezing a subsequent reproduction does not remove the earlier selection.

The inclusive family has 20 successful assignments. This is a separate sensitivity result, not a favorable replacement for the primary calibration. Neither fraction adjusts for all earlier trait families or the restriction to the eleven candidates. Neither is the probability that selection was deliberate, that the pattern is a historical coincidence, or that a trapdoor exists.

The counts are exact for this census. Uncertainty concerns cohort ascertainment, historical label interpretation, unrecorded research choices and external generalization. There is no untouched historical holdout in this study.

7 Generator and coefficient comparisons

The strict statistic depends only on p, n. A different generator, coordinate representation or $\mathbb{F}_p$-isomorphic model preserves them and hence the profile. For a fixed field this is ultimately a property of the point count, not a recovered formula for an unknown generator.

The earlier generator/coefficient census [6] checked 100,548 point-trait cells under repeated model and point controls, and 147,875 same-field difference cells. Its 19,152 single-envelope tests gave no complete separator; 880 public-scalar equality checks gave no matches. These finite negatives do not exclude every multivariate rule or generator relation. The eleven historical unpublished generators remain unknown.

For $y^2 = x^3 - 3x + b$, direct substitution gives

$$x = 0 \Longrightarrow y^2 = b, \qquad x = 1 \Longrightarrow y^2 = b - 2 \pmod{p}. \tag{6}$$

These defining-equation identities are not additional evidence of hidden selection. Earlier conditional Lean proofs cover the identities and seed-equation substitutions, with no authentication of historical generator choices. The current census has native exact verification, not a new Lean formalization. No new algebraic theorem is asserted by this manuscript.

8 Historical and security interpretation

Frobenius splitting matters in point-counting algorithms [8], providing a mathematical reason to examine this pattern in a generation setting. The count alone does not measure an implementation's point-counting time, explain a preference for lower counts or identify the original acceptance

predicate. Earlier inclusive counts and product-coverage experiments use different statistics and retain their counterexamples.

A historical explanation needs an authenticated implementation or decision trace fixing seed consumption, coefficient choices, actual point-count inputs and keep/reject outcomes, or independently specified predictions beyond this fitted cohort. The source audit [5] and public accounts [2, 9] supply no such trace. Reconstructed eligibility is not observed visitation.

No discrete-logarithm algorithm follows from the separation. An isogeny can transport a group problem without providing a cheaper destination solver. An advantage requires explicit maps, subgroup behavior, a destination algorithm and complete costs. Different point counts also preclude interpreting the characters as isogeny maps between the compared published and unpublished rows over the same field. We construct no maps, recover no unknown scalars and provide no curve-specific security certification. Neither weak publications nor weak unpublished candidates are established.

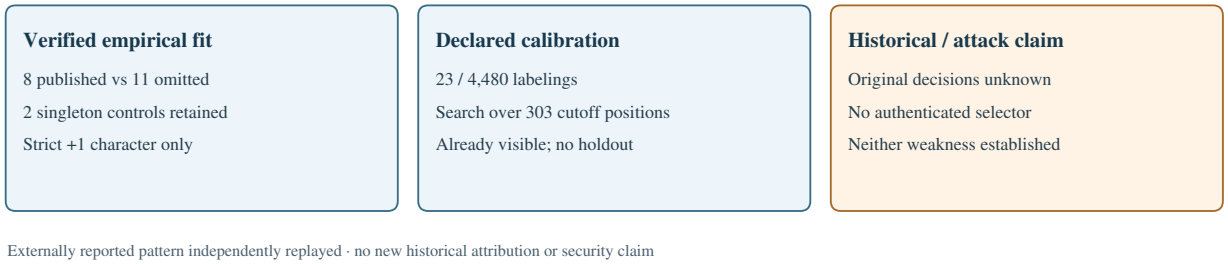


Figure 2: Evidence levels, reusing the frozen study's diagram. Exact numerical conclusions do not supply the missing historical or attack mechanism.

9 Reproducibility and data availability

The package contains byte-preserved inputs, the 21-row table, native summary and verification metadata, scientific source copies, editable vector figures and LaTeX source. `data/source-bindings.json` records source paths and digests. The source snapshot is `c51b0c32620ab43c73b1c0a34c236412f466f89a`; the actual execution and preparation commits are separately fixed in the original protocol.

The full run archive is addressed by SHA-256

`666660f97e6a7e2e9693a74789b39ae25e11a4baddbcae031d2fd02e3b7ac899`.

Its 96 regular members include native evidence and refusal controls. The delivery record reports a full remote download and all-member verification. Preparation/document archives, failed planning records, binary bindings, protocol and limits remain preserved. Library versions are pinned, not bundled. This manuscript adds no scientific rerun, factorization or point-count certificate. Typesetting checks do not constitute cryptographic replay.

The README explains PDF compilation, evidence bindings and access to the original canonical export and complete raw record. The wider arithmetic and generator evidence retains its original provenance [6, 4]; indeterminate historical-tool results remain indeterminate.

10 Conclusion

The prior strict splitting-count separation reproduces exactly on its fixed 21-choice cohort, with eight publications and eleven unpublished candidates in paired fields. Separate implementations and deliberate corruption controls support the finite result. Its dependence on excluding zero characters is visible in P-192; its calibration covers only a declared cutoff family. The historical choice mechanism and both weakness hypotheses remain open. Independently specified testing on a justified expanded cohort, or authenticated original decision evidence, would discriminate further.

Draft and assistance statement. This anonymous replication manuscript was prepared with AI assistance in drafting and document production from the cited computation records. Human authorship, scientific review and final submission metadata remain to be supplied. No IACR acceptance or peer-review status is claimed.

A Every recorded choice at cutoff 691

The table retains every choice, including both singleton publications. Names are historical publication names; exact parameters and stream identities are in `data/inputs.tsv`.

Table 3: All native observations. P denotes published; U denotes unpublished. Source: `data/at691.tsv`.

Record / historical name	Label	S_{691}	R_{691}	I_{691}
`p192-139` / NIST P-192	P	60	3	63
`p192-2198`	U	61	0	61
`p192-5228` / prime192v3	P	57	1	58
`p192-5405`	U	62	0	62
`p192-6013` / prime192v2	P	58	1	59
`p224-777`	U	53	4	57
`p224-1485`	U	56	3	59
`p224-5736`	U	59	2	61
`p224-7026` / NIST P-224	P	50	3	53
`p239-36`	U	71	2	73
`p239-1562`	U	67	3	70
`p239-2185` / prime239v1	P	42	1	43
`p239-2320` / prime239v2	P	57	2	59
`p239-2491`	U	63	2	65
`p239-2682`	U	61	4	65
`p239-2863`	U	66	1	67
`p239-3048` / prime239v3	P	57	1	58
`p256-876`	U	58	2	60
`p256-1842` / NIST P-256	P	57	2	59
`p384-2688` / NIST P-384	P	54	0	54
`p521-1136` / NIST P-521	P	52	1	53

B Evidence map

Paths are relative to the cited repository snapshot.

- Statistic, exact inputs, sources, protocol and results: `crypto:research/nist_skip_criterion_20261025/strict_elkies_20261029/`.
- Generator/coefficient census: `crypto:research/nist_skip_criterion_20261025/generator_b_execution_20261029/`.
- Conditional identity proofs: `crypto:research/nist_skip_criterion_20261025/generator_b_20261029/lean/CurveEquation.lean`.
- Draft and expanded-cohort audits: `crypto:research/nist_skip_criterion_20261025/source_check_20261027/sources/`.
- Proved-factor follow-up: `crypto:research/nist_skip_criterion_20261025/factor_followup_20261026/`.

The computational claims are finite observations with separate native verification. The splitting interpretation is standard background [8]; the generator identities have earlier conditional Lean proofs. Historical intent, exhaustive coverage of mathematical traits and any solver advantage have no supplied proof.

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